

Study on Establishing Structured Born-Digital Online Co-Authoring System

Hyun Ji*, Wan Jong Kim**

Korea Institute of Science Technology and Information

E-mail: ji_hyun@kisti.re.kr*, wjkim@kisti.re.kr**

1. Introduction

An increasing number of researchers are opting to cooperate while conducting research and writing articles. According to the Korea Citation Index (KCI), in 2021, a total of 114,049 articles were written and published by 305,602 authors [1]. Alternatively, an article was written by 2.67 authors on average, and this number increased post 1990s. Additionally, interdisciplinary research is being conducted actively in various fields. Moreover, the increase in the use of the Internet affects the entire field of scholarly communication. The researchers search related articles from e-journals, submit drafts to online peer review/publishing systems, and share ideas on various platforms. It makes the researchers cooperate actively and digital publishing more essential; however, it causes an increase in the cost of publishing. Therefore, it is necessary to realize open collaboration and open access in the internet environment.

In this context, to support online co-authoring and publishing, a structured born-digital online co-authoring system was established. The system is based on Collab, which is an online co-authoring system from AccessON, the national Open Access Platform of Korea. The system facilitates article creation in a structured and machine-readable format. It includes a co-authoring system and MS Word to Journal Article Suites (JATS) XML transition system.

2. Co-authoring System

The co-authoring system includes various services from Collab to support each step of the authoring process. Collab supports the entire process of article creation, from searching for references to publishing the article using three tools: Collab Gate, Collab Platform, and Collab Pen.

In the first step, namely, research, researchers search related articles; this step is supported by Collab Gate. Collab Gate is a Chrome extension which identifies whether the searched paper is an open access paper and saves the papers projects or My Documents on the Collab Platform. The Collab platform allows researchers to manage an ongoing article, references, and various files under a project and invite coauthors to contribute to the project.

While authoring an article, every author who is invited to the project can create content simultaneously by using Collab Pen, an MS Word plug-in service. Similar to reference management tools such as RefWorks, Collab Pen offers a function to insert citations and images. However, different from commercial tools, Collab Pen provides integration services with AccessON and Acoms+, i.e., AccessON Peer Review Management System Plus. The author, by using the integration service can send a completed article to the self-archiving system or peer review system without leaving the application. Additionally, the integration system offers a metadata autocomplete service during self-archiving or submission. At the final step, namely, publishing, Collab Pen supports the JATS XML transition based on an MS Word template with the Styles menu.

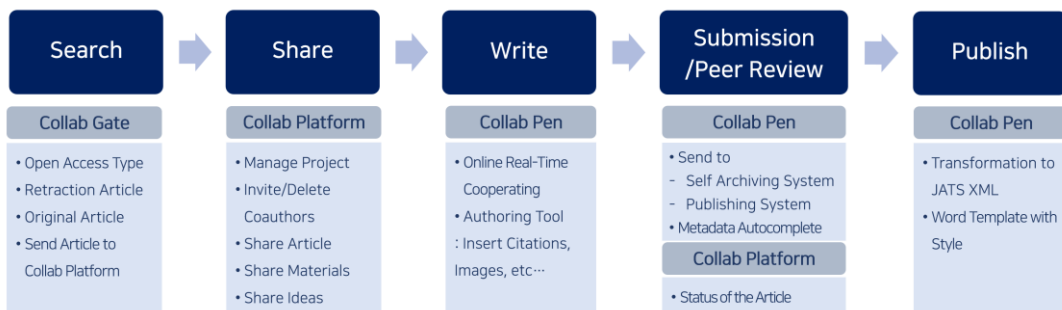


Figure 1 Online Born-Digital Co-authoring System

3. MS Word to JATS XML Transition System

JATS defines a set of XML elements and attributes for tagging journal articles and describes three article models. One of the biggest merits of JATS XML is that it offers high quality online publishing and distribution through journal sites [2]. However, transition to JATS XML is difficult and cost intensive. Before such transition, every element of an article, such as the main text and references should be distinguished and separated manually to match the exact tag of JATS XML. This process is time, labor, and cost intensive. This is the reason why the scholarly society faces challenges in publishing articles online.

To address this problem, Collab offers MS Word templates with the Styles menu, which is a set of formatting instructions for article elements such as Title_AccessON_02. By applying a style, the elements are separated automatically, which makes the JATS XML transition more cost-effective. During the JATS XML transition process (Figure 2), error identification and revision guide were provided so that the writer could accurately apply styles.

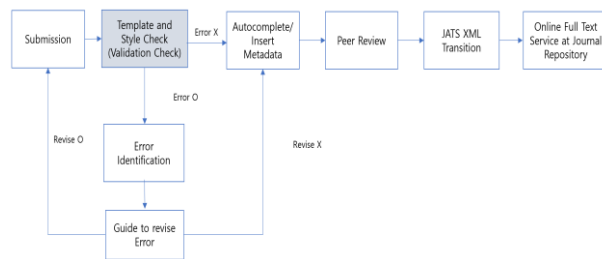


Figure 2 JATS XML Transition Process

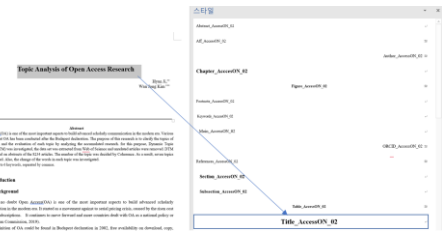


Figure 3 Style Application Example (Title Style)

4. Evaluation of Born-Digital Online Co-Authoring System

As a part of the establishment process, an evaluation of the system including the JATS XML transition program was conducted by nine professionals involved in online journal publishing, such as an editor-in-chief (EIC) of a scholarly journal. Both quantitative and qualitative evaluations were conducted on the ten questions categorized in (1) the quality of the transmitted JATS XML file, (2) completeness of the system, and (3) usability of the system. A total of ten points were allotted to each question, and for the quantitative evaluation, a 7-point Likert scale was used. The evaluation revealed that the average of the total score except for minimum and maximum scores was 81.43.

In the qualitative evaluation, evaluators provided suggestions for improving the system, such as a guide for the editor and writer. After the evaluation, the system was updated, based on the provided suggestions.

Table 1 Result of Evaluation

	Question/Evaluator	A	B	C	D	E	D	E	F	G
1 (30)	Are tags and contents of the transitioned JATS XML file matched?	8.33	10.00	8.33	6.67	8.33	6.67	8.33	8.33	10.00
	Are tables transitioned accurately?	6.67	10.00	8.33	6.67	8.33	8.33	10.00	10.00	10.00
	Are figures transitioned accurately?	5.00	10.00	8.33	6.67	8.33	8.33	6.67	8.33	10.00
2 (30)	Are the steps of the co-authoring system matched with the real research process?	8.33	10.00	8.33	5.00	8.33	8.33	8.33	6.67	10.00
	Are the services of the co-authoring process useful for each step?	8.33	5.00	8.33	5.00	8.33	10.00	10.00	6.67	10.00
	Are the services inclusive for online authoring?	10.00	8.33	8.33	6.67	8.33	8.33	8.33	8.33	10.00
3 (40)	Are the services including template download convenient to use?	5.00	8.33	8.33	6.67	8.33	8.33	10.00	8.33	8.33
	Does the process provide sufficient validation check steps?	6.67	10.00	8.33	5.00	8.33	8.33	8.33	6.67	8.33
	Does the process provide sufficient revision steps?	5.00	10.00	8.33	5.00	8.33	6.67	6.67	8.33	8.33
	Does the system satisfactorily support online publishing?	8.33	5.00	6.67	6.67	8.33	8.33	8.33	8.33	8.33
TOTAL		71.67	86.67	81.67	60.00	83.33	81.67	85.00	80.00	93.33

5. Conclusion

To realize open access in the digital domain, it is essential to support both co-authoring and online publishing because an increasing number of researchers are cooperating and conducting research online. The structured Born-Digital Online Co-Authoring System proposed in this study can provide a solution to achieve the same.

Acknowledgments

This research is part of the research project titled "Establishment of open access support system of science and technology information through digital transformation" supported by the Korea Institute of Science and Technology Information.

References

- [1] KCI, <https://www.kci.go.kr/>
- [2] Huh, S., "Journal Article Tag Suite 1.0: National Information Standards Organization standard of journal extensible markup language," Sci Ed, Korean Council of Science Editors, Korea, 2014, pp. 99–104.